Notes: Klingler and Lando (RFS, 2018)

Safe Haven CDS Premiums

Contents

1 Executive overview

1.1 Empirical puzzle

In a frictionless setting (no-arbitrage, no frictions), the CDS premium on an entity and the corresponding cash-bond yield spread should be tightly linked (the "CDS–bond basis" should be small once we adjust for matching maturities, funding, and other technicalities). The paper documents that this link *breaks down sharply for safe sovereigns*:

- For safe-haven sovereigns (e.g., Germany, U.S., Japan, U.K.), changes in CDS premiums are almost unrelated to changes in bond yields/spreads.
- For riskier sovereigns (e.g., Italy, Spain, Portugal), the CDS–yield link is much closer to the textbook relationship.

1.2 Core mechanism: CDS as regulatory capital relief (CVA charge channel)

The central hypothesis is that CDS on sovereigns is not priced only as default insurance. Instead, CDS can be used by derivatives-dealing banks to *reduce Basel III capital requirements* arising from uncollateralized OTC derivatives with sovereigns:

1. Many sovereigns do *not* post collateral on OTC derivatives (interest-rate swaps, etc.). Banks therefore face counterparty-credit exposure.
2. Basel III introduced a capital charge for *CVA risk* (mark-to-market losses from *deterioration* in counterparty credit quality, not just default).
3. The regulatory calculation uses CDS levels and CDS volatility; hence even a "safe" sovereign with a small but positive CDS premium can generate meaningful CVA capital charges.
4. Banks can obtain capital relief by purchasing CDS protection on the counterparty. This creates *inelastic demand* for CDS tied to derivative exposures.

1.3 Main conclusion

A non-trivial part of sovereign CDS premiums—especially for safe sovereigns—reflects a *regulatory premium* (the value of capital relief), not only credit risk. The paper supports this with:

- a tractable equilibrium model with margin and capital constraints,
- evidence that dealer banks are net buyers of sovereign CDS,
- a strong relationship between sovereign CDS volumes and banks' derivatives exposures toward sovereigns,
- weak co-movement of CDS and yields for safe sovereigns, and
- additional tests using regulatory proxies and corporate bond/CDS evidence.

2 Institutional background: CVA capital and why sovereign CDS is special

2.1 Key facts

- **Uncollateralized sovereign OTC.** Sovereigns typically do not post collateral in OTC derivatives, unlike most financial counterparties. This makes counterparty risk salient for banks.
- **CVA is about credit deterioration.** CVA increases when counterparty credit quality worsens (CDS spreads rise), generating mark-to-market losses on the bank's derivatives book.
- **Basel III CVA capital charge.** Banks must hold capital against *changes* in CVA (CVA VaR). The required capital increases with CDS spread *level* and *volatility*.
- **CDS hedge provides capital relief.** Buying CDS protection on the counterparty reduces or removes this charge, creating demand for CDS even when expected default losses are tiny.

2.2 Empirical implication

For safe sovereigns, the cash bond yield may be dominated by risk-free rates and safe-asset convenience effects, while CDS pricing can reflect a separate demand component coming from regulatory hedging.

3 The model (one-period equilibrium with capital relief)

3.1 Assets

There are three assets:

1. **Risky asset.** Price normalized to 1. Time-1 payoff
$$\tilde{r} \sim \mathcal{N}(1+\mu, \sigma^2).$$
A position of dollar size g requires margin $m|g|$. The parameter m also proxies how much the risky asset contributes to a bank's risk-weighted assets (a reduced-form "balance sheet tightness").

2. **Risk-free asset.** Gross return $1 + r$.

3. **CDS on a (safe) sovereign.** Premium per unit notional is s (endogenous). From the protection buyer's perspective, the CDS payoff is
$$\tilde{s} = \begin{cases} -s, & \text{w.p. } 1-p, \\ \mathrm{LGD}, & \text{w.p. } p, \end{cases} \qquad \Rightarrow \qquad \bar{s} := \mathbb{E}[\tilde{s}] = p\,\mathrm{LGD} - (1-p)s.$$
Initial margin is n^+ for buying protection and n^- for selling protection (both per unit notional).

3.2 Agents and wealth

There are two agents:

- B: a derivatives-dealing bank with an uncollateralized OTC exposure to the sovereign.
- E: an end user who can sell CDS protection.

If agent $i \in \{B, E\}$ invests $g \in \{b, e\}$ dollars in the risky asset and takes CDS notional position $\bar{g} \in \{\bar{b}, \bar{e}\}$ (positive for buying protection, negative for selling), then terminal wealth is:

$$W_1^i = W_0^i(1+r) + g(\tilde{r} - r) + \bar{g}\,\tilde{s}.$$

3.3 Preferences and objective

Agents have mean-variance preferences. They solve:

$$\max_{g,\bar{g}} \left[g(\mu - r) + \bar{g}\,\bar{s} - \frac{1}{2}(\sigma g)^2 - \frac{1}{2}v(s)\bar{g}^2 \right], \tag{1}$$

where the variance of the CDS payoff is approximated as

$$v(s) = (p - p^2)\left(\mathrm{LGD}^2 + 2s\,\mathrm{LGD}\right).$$

(They drop the $(p-p^2)s^2$ term since it is small for the relevant range of CDS premiums.)

3.4 Constraints

3.4.1 End user's margin constraint

Selling CDS requires initial margin. The end user faces:

$$me + n^-|\bar{e}| \le W_0^E. \tag{2}$$

In equilibrium the end user sells protection, so $\bar{e} < 0$ and $|\bar{e}| = -\bar{e}$.

3.4.2 Bank's capital relief constraint

The bank has a legacy uncollateralized derivatives position with *expected exposure* EE. This creates a capital requirement proportional to EE. Buying CDS notional $\bar{b}$ (up to EE) provides capital relief. The bank faces:

$$mb + n^+\bar{b} + \kappa(EE - \bar{b}) \le W_0^B, \tag{3}$$

$$\bar{b} \le EE. \tag{4}$$

Here κ is the *capital charge per unit exposure* (a reduced-form summary of Basel III CVA capital).

3.5 Market clearing

Only B and E trade CDS, so:

$$\bar{b} + \bar{e} = 0. \tag{5}$$

4 Solving the model: equilibrium CDS premium and interpretation

4.1 Parameter restrictions used for a clean result

The paper imposes three inequalities to focus on the empirically relevant constrained region:

$$\frac{\mu - r}{\sigma^2} > \frac{1}{m} \max\left\{W_0^E,\ W_0^B - n^+ EE\right\}, \tag{6}$$

$$\min\left\{\frac{W_0^E}{n^-},\ \frac{W_0^B}{\kappa}\right\} > EE, \tag{7}$$

$$\kappa > n^+. \tag{8}$$

Intuition:

- (6): both agents are constrained (they would like to lever more into the risky asset than allowed).
- (7): both sides have enough wealth to trade CDS at notional EE (full hedging feasible).
- (8): capital relief dominates the margin cost for buying CDS, so the bank gains "investable slack" from hedging.

4.2 Proposition 1 (full-protection equilibrium)

Define $R := p \cdot EE \cdot \mathrm{LGD}$. Then define:

$$s_b := \frac{1}{(1-p)(1+2R)} \left[\frac{\kappa - n^+}{m}\left(\mu - r - \frac{\sigma^2}{m}\left(W_0^B - EE\, n^+\right)\right) + p\,\mathrm{LGD}\right] - \frac{R\,\mathrm{LGD}}{1+2R}, \tag{9}$$

$$s_f^E := \frac{1}{(1-p)(1-2R)} \left[\frac{n^-}{m}\left(\mu - r - \frac{\sigma^2}{m}\left(W_0^E - n^- EE\right)\right) + p\,\mathrm{LGD}\right] + \frac{R\,\mathrm{LGD}}{1-2R}. \tag{10}$$

Proposition. If $s_f^E \leq s_b$, then the unique equilibrium CDS premium is $s = s_f^E$ and the bank buys full protection $\bar{b} = EE$.

4.3 Economic interpretation

- s_b is the *bank's cutoff willingness-to-pay* for keeping full insurance demand $\bar{b} = EE$.
- s_f^E is the *end user's price* at which she is willing (given her own margin constraint) to supply exactly EE notional.
- If the end user can supply EE at a premium that is still acceptable for the bank ($s_f^E \leq s_b$), then the equilibrium occurs in the region where bank demand is effectively *inelastic* at $\bar{b} = EE$.

4.4 Why safe entities can have strictly positive CDS premiums

A key qualitative implication follows from (10):

- Even as $p \to 0$ (default probability tends to zero), the CDS premium need not go to zero.
- The premium contains a component that compensates sellers for margin usage and reflects the bank's willingness to pay for capital relief.

Hence, for safe sovereigns, *credit risk can be tiny while the CDS premium remains economically meaningful.*

4.5 Numerical example (Figure 5)

The paper illustrates supply (end user) and demand (bank) in Figure 5. The demand curve is flat at $\bar{b} = EE$ for a range of CDS premiums (full hedging region), while supply is increasing in s after a threshold. The equilibrium premium can substantially exceed the "risk-neutral" credit-risk-only value.

5 Proof outline of Proposition 1 (Appendix B, re-derived)

The proof uses Kuhn–Tucker conditions to solve both agents' constrained mean-variance problems.

5.1 End user

To avoid negative choice variables, let $\bar{e} \geq 0$ denote the number of CDS contracts *sold* by the end user. Then the end user's Lagrangian is:

$$\mathcal{L}_E(e, \bar{e}, \lambda) = e(\mu - r) - \bar{s}\,\bar{e} - \frac{1}{2}(\sigma e)^2 - \frac{1}{2}v(s)\bar{e}^2 - \lambda\left(me + n^-\bar{e} - W_0^E\right). \tag{11}$$

KT conditions (with complementary slackness) yield, in the constrained case $\lambda > 0$,

$$e = \frac{\mu - r - \lambda m}{\sigma^2} =: e^U - \lambda\frac{m}{\sigma^2}, \tag{12}$$

$$\bar{e} = \frac{-\bar{s} + \lambda n^-}{v(s)} =: \bar{e}^U - \lambda\frac{n^-}{v(s)}, \tag{13}$$

$$\lambda = \frac{me^U + n^-\bar{e}^U - W_0^E}{C_E(s)}, \qquad C_E(s) = \frac{m^2}{\sigma^2} + \frac{(n^-)^2}{v(s)}. \tag{14}$$

A supply threshold exists: the end user supplies CDS only if the premium is high enough so that selling protection has positive expected return (equivalently, $\bar{s} < 0$).

5.2 Bank

In the region where the bank buys full protection ($\bar{b} = EE$), the constrained optimum satisfies:

$$\bar{b} = EE, \qquad b = b^U - \lambda_1\frac{m}{\sigma^2},$$

and the condition for staying in the full-protection corner is $\lambda_2 > 0$, which implies an upper bound $s < s_b$ (the s_b in (9)).

5.3 Equilibrium

If supply reaches $\bar{e} = EE$ at a premium below s_b, then market clearing pins down the unique equilibrium premium as $s = s_f^E$ (the s_f^E in (10)) and $\bar{b} = EE$.

6 From regulation to the reduced-form parameter $\kappa(s)$ (Appendix C)

6.1 CVA definition

Basel defines the CVA of a bank's derivatives position with a risky counterparty as

$$\text{CVA} = \text{LGD} \sum_{i=1}^{T} Q(\tau \in (t_{i-1}, t_i)) \; EE(t_{i-1}, t_i), \tag{15}$$

where τ is the default time and $Q(\cdot)$ is the risk-neutral default probability inferred from CDS premiums.

6.2 CVA VaR and capital

A simplified CVA VaR formula used in the paper is:

$$\text{CVA\_VaR} = 3 \times \text{WorstCase} \times \text{CS01}, \tag{16}$$

$$\text{WorstCase} = \sigma_{\text{CDS}} \sqrt{\frac{10}{252}} \Phi^{-1}(0.99), \tag{17}$$

$$\text{RWA} = 12.5 \left(\text{CVA\_VaR} + \text{CVA\_StressedVaR}\right). \tag{18}$$

Here CS01 is the "credit delta": sensitivity of CVA to a 1bp CDS move. Under a flat CDS term structure and constant EE,

$$\text{CS01} = EE \cdot 10^{-4} \sum_{i=1}^{T} \left(t_i e^{-st_i/\text{LGD}} - t_{i-1} e^{-st_{i-1}/\text{LGD}} \right) \frac{D_{i-1} + D_i}{2}, \tag{19}$$

where D_i are discount factors.

6.3 Mapping into the model: $\kappa(s)$

If capital is set to $0.1 \cdot \text{RWA}$, the per-unit-exposure capital charge in the model becomes

$$\kappa(s) = 0.1 \cdot 12.5 \cdot c \cdot \frac{\text{CS01}}{EE} \left(\sigma_1^{(st)} + \sigma_3^{(st)} \right), \tag{20}$$

where $\sigma_1^{(st)}$ is a one-year CDS volatility, $\sigma_3^{(st)}$ is a stressed (max) volatility over the last three years, and c collects constants from (16)–(17). This shows why **CDS volatility and the CDS level enter regulatory demand and thus equilibrium CDS premiums**.

7 Empirical strategy and evidence: interpreting each table and figure

7.1 Data map (what each block of evidence does)

The empirical analysis has four blocks:

1. **Volumes:** Are CDS volumes related to banks' derivatives exposures toward sovereigns?

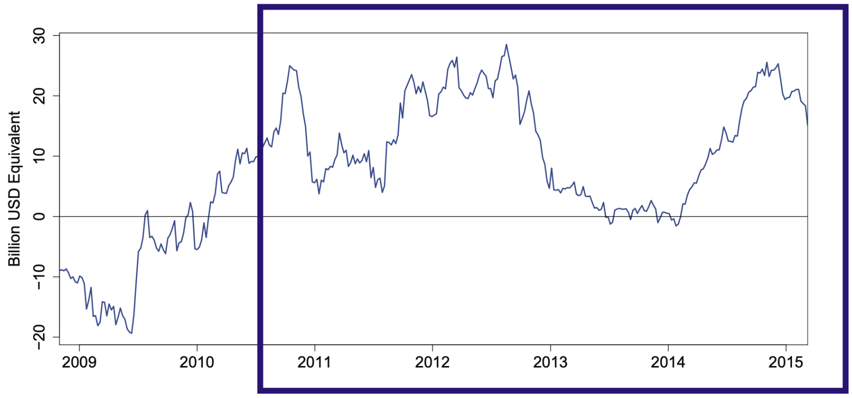


Figure 3
Derivatives dealers are net buyers of sovereign CDS

2. **Prices I:** Are changes in CDS premiums linked to changes in bond yields, and does this depend on sovereign safety?

3. **Prices II:** What variables explain CDS premium changes (credit-risk proxies vs regulatory proxies)?

4. **External validity:** Does the same "safe-entity disconnect" appear for safe corporate issuers, especially those that do not collateralize derivatives?

7.2 Figure 3: dealers are net buyers of sovereign CDS

What it plots. The difference between gross sovereign CDS notional where derivatives dealers buy protection and where they sell protection (DTCC).

Interpretation.

- Post-2010, dealer banks become *net buyers* of sovereign CDS.
- This is consistent with CDS being used to hedge (and obtain capital relief from) uncollateralized sovereign derivative exposures.

7.3 Figure 4: who collateralizes derivatives?

What it plots. For major U.S. dealers, the fraction of derivatives exposure that is uncollateralized, by counterparty type: sovereign, corporate, financial.

Interpretation.

- Sovereign exposures are almost fully uncollateralized (fraction near 1).
- Corporate exposures are largely uncollateralized (substantial fraction).
- Financial exposures are close to fully collateralized (fraction near 0).

This directly motivates the later corporate tests: if the CVA channel requires uncollateralized OTC exposure, it should matter more for nonfinancial corporates than for financials.

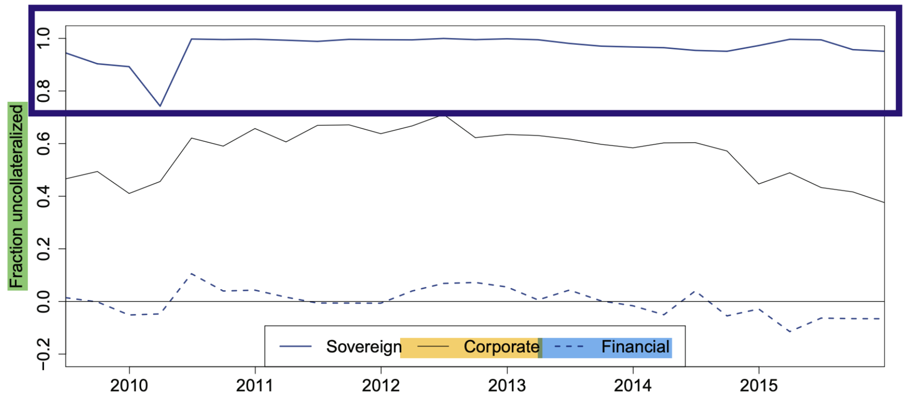


Figure 4
Fraction of bank's uncollateralized derivative positions

Table 1
CVA calculations based on EBA stress tests

	Mio USD			Basis points				
	Notional value	Fair value	CDS outst	CDS prem	$\sigma_1(s_t)$	$\sigma_3(s_t)$	$\frac{CS01}{EE}$	$\kappa(s)$
GER	402,855	34,072	13,118	42	19	24	0.26%	0.150
AUT	28,403	1,644	4,224	44	40	49	0.16%	0.271
FIN	95,414	5,073	2,189	30	14	18	0.13%	0.087
FRA	47,938	3,210	11,742	92	46	55	0.14%	0.234
ITA	106,959	19,136	16,916	284	118	133	0.32%	0.495
POR	9,423	564	3,684	430	214	290	0.09%	0.821
ESP	27,691	1,883	9,259	291	118	123	0.10%	0.401
GBR	7,920	19,255	5,842	42	12	30	3.08%	0.072
JAP	17,471	5,269	9,189	81	20	31	0.63%	0.091
USA	77,995	54,710	3,389	36	10	19	1.56%	0.052

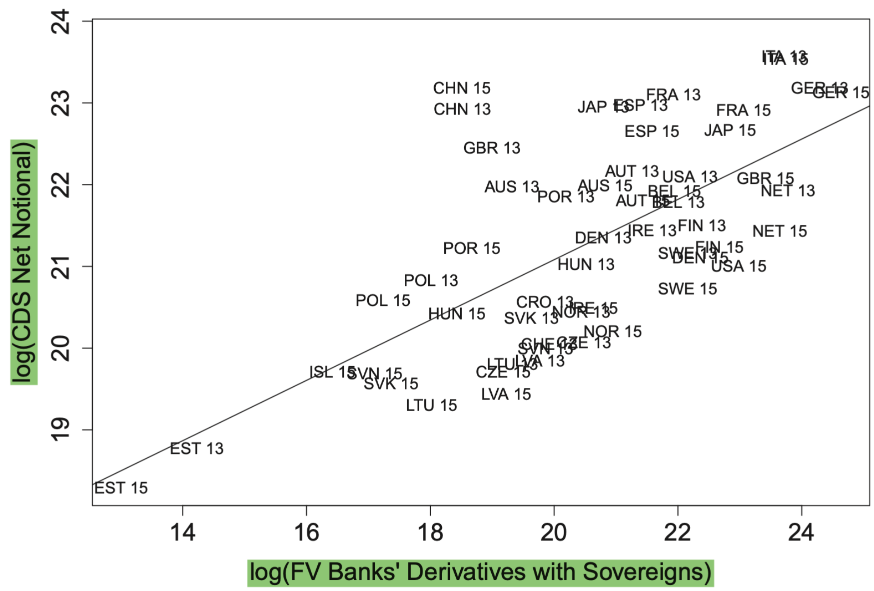


Figure 6
Banks' derivatives exposures and CDS volumes outstanding

7.4 Table 1: CVA calculations and the size of capital relief

Object. Table 1 combines EBA stress-test derivatives data (notional and fair value) with CDS premiums and volatility measures to compute objects relevant for Basel III CVA capital.

Key columns and meaning.

- *Fair value (positive)* is used as a proxy for EE (expected exposure).
- $\sigma_1^{(st)}$ and $\sigma_3^{(st)}$ enter the regulatory CVA VaR.
- $\kappa(s)$ (right-most column) is the implied capital charge per unit exposure (model parameter).

Interpretation.

- The computed $\kappa(s)$ is economically large, implying that buying CDS can free enough regulatory capital to justify paying CDS premiums even for safe sovereigns.
- This is a necessary "reality check": if $\kappa(s)$ were negligible, the capital-relief mechanism would not plausibly move prices.

7.5 Figure 6 and Table 2: CDS volumes scale with derivatives exposures

Figure 6. Scatter of log(CDS net notional) on log(fair value of banks' sovereign derivatives) across countries and years (2013/2015). The fitted line is upward sloping.

Table 2 (Equation (10)). Cross-sectional regression:

$$\log(\text{CDS}_{i,t}) = \alpha_0 + \alpha_1 \mathbf{1}\{t = 2015\} + (\beta_{FV,0} + \beta_{FV,1}\mathbf{1}\{t = 2015\}) \log(\text{FV}_{i,t})$$
$$+ (\beta_{Debt,0} + \beta_{Debt,1}\mathbf{1}\{t = 2015\}) \log(\text{Debt}_{i,t}) + \varepsilon_{i,t}.$$

Interpretation of Table 2.

Table 2
Banks' derivatives exposures and CDS volumes outstanding

	(1)	(2)	(3)	(4)
Intercept	13.70***	13.73***	6.96***	6.58***
	(0.89)	(1.65)	(1.41)	(1.59)
Intercept $\times \mathbb{1}_{\{2015\}}$		−0.05		0.81
		(1.95)		(2.96)
$\log(FV)$	0.37***	0.37***	0.10***	0.11**
	(0.04)	(0.08)	(0.04)	(0.05)
$\log(FV) \times \mathbb{1}_{\{2015\}}$		−0.01		0.00
		(0.09)		(0.09)
$\log(Debt)$			0.46***	0.48***
			(0.07)	(0.07)
$\log(Debt) \times \mathbb{1}_{\{2015\}}$				−0.04
				(0.16)
Observations	55	55	55	55
Adjusted R^2	0.45	0.44	0.77	0.76

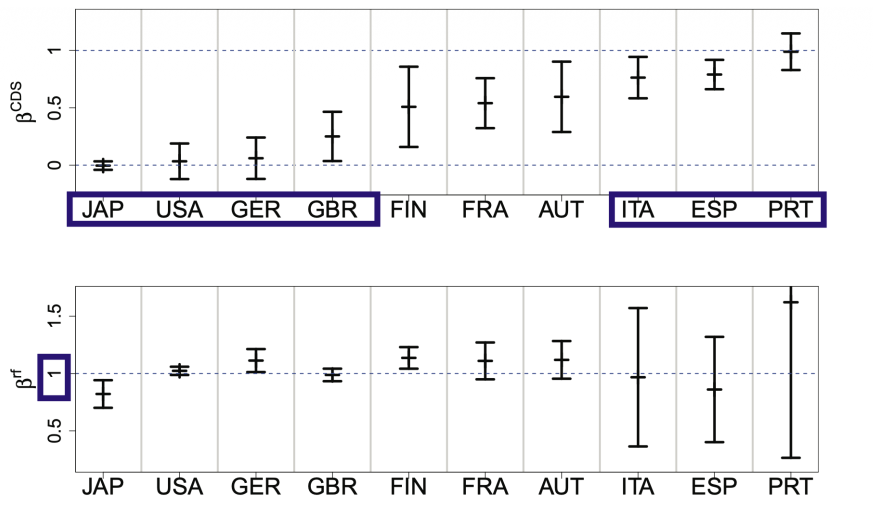


Figure 1
Explaining bond yields with risk-free rates and credit risk

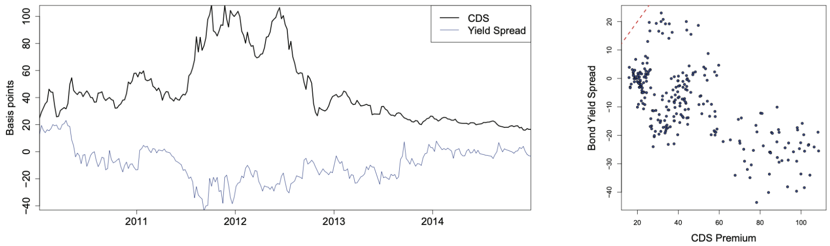
- log(FV) enters positively and strongly: sovereigns with larger banks' derivatives exposures have larger CDS volumes outstanding.
- Adding log(Debt) increases explanatory power: debt size also correlates with CDS volumes, but the derivatives-exposure measure remains significant.
- Interactions with the 2015 dummy are small: the relationship is stable across the two stress-test snapshots.

7.6 Figure 1 and Equation (11): bond yields respond weakly to CDS for safe sovereigns

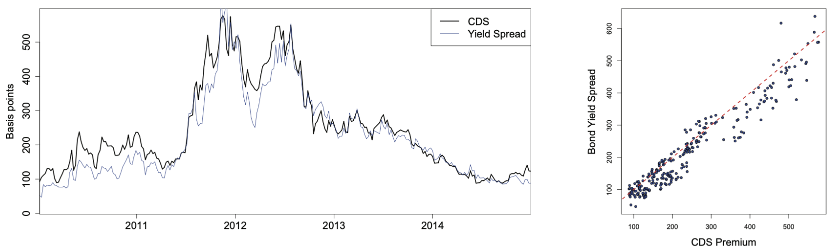
The core regression used to link yield changes to CDS changes is:

$$\Delta \text{Yield}_{i,t} = \alpha + \beta^{CDS} \Delta \text{CDS}_{i,t} + \beta^{rf} \Delta r^{f}_{i,t} + \varepsilon_{i,t}. \tag{21}$$

Figure 1. Shows β^{CDS} and β^{rf} (with confidence intervals) across ten sovereigns:



A. CDS premium and bond yield for Germany



B. CDS premium and bond yield for Italy

Figure 2
The disconnect between CDS premiums and bond yield spreads

- For safe havens (Japan, U.S., Germany), β^{CDS} is near zero: yield changes are not explained by CDS changes.
- For risky sovereigns (Italy, Spain, Portugal), β^{CDS} is much closer to one.
- β^{rf} is close to one across countries, consistent with the swap rate proxy capturing risk-free-rate movements.

7.7 Figure 2: a transparent visual of the disconnect

Germany vs Italy.

- Germany: CDS and yield spreads show weak co-movement; scatter plot suggests little (or even negative) relationship.
- Italy: CDS and yield spreads move closely together; scatter plot shows a strong positive relationship.

This is the graphical motivation for the cross-sectional pattern emphasized throughout the paper.

7.8 Table 3: ruling out alternative explanations for safe-haven yields

Regression. For safe sovereigns, Table 3 estimates:

$$\Delta\text{Yield}_t = \alpha + \beta_{CDS}\Delta\text{CDS}_t + \beta_{rf}\Delta r_t^f + \beta_{CY}\Delta CY_t + \text{Controls}_t + \varepsilon_t.$$

Main interpretation.

- β_{rf} is large and significant (close to one): safe-haven yields move with risk-free rates.
- β_{CDS} is small/insignificant for Japan and the U.S.: CDS changes are not pricing into yields.

Table 3
Explaining bond yields with credit risk, risk-free rates, and convenience yield

	Japan	U.S.	Germany	U.K.
Intercept	−0.19*	0.04	−0.21	0.10
	(0.11)	(0.13)	(0.26)	(0.35)
ΔCDS_t	−0.01	−0.00	0.08	0.20*
	(0.02)	(0.07)	(0.09)	(0.12)
Δrf_t	0.82***	0.89***	0.87***	0.75***
	(0.06)	(0.05)	(0.06)	(0.06)
ΔCY_t	−0.07	0.03	−0.07	−0.15
	(0.11)	(0.11)	(0.06)	(0.11)
$\Delta OnOff_t$		−0.33**		
		(0.15)		
$\Delta Turnover_t$		0.99		0.62
		(0.67)		(0.73)
ΔKfW_t			−0.23***	
			(0.06)	
ΔCtD_t	0.17	1.99***	3.72***	4.25***
	(0.13)	(0.60)	(0.83)	(0.74)
Observations	240	253	252	170
Adjusted R^2	0.73	0.96	0.84	0.86

- Additional controls for convenience yield and liquidity (on/off-the-run spreads, turnover, KfW spread) do not restore a strong CDS–yield link.
- The CtD proxy enters significantly for several countries, suggesting CDS contract design features matter, but it does not overturn the core "disconnect" fact.

7.9 Table 4: extended cross-section (23 countries, monthly 10-year data)

Table 4 provides two complementary views.

7.9.1 Panel A: yields on CDS (and the role of dealer-bank buying pressure)

Regression idea. Panel A estimates (21) on a broad cross-section, separately for risk categories, and includes an interaction with an indicator for unusually large increases in dealer banks' CDS positions.

Interpretation.

- The slope ΔYield on ΔCDS is:
  - close to zero (even negative) for *safe* countries,
  - moderate for *low risk* countries,
  - large for *risky* countries.
- The interaction term is negative for low-risk and risky groups: when dealer banks increase CDS positions strongly, the CDS–yield link weakens.

7.9.2 Panel B: CDS on yield spreads and volatility proxies

Regression (Equation (13)).

$$\Delta \text{CDS}_{i,t} = \alpha + \beta_{YS}\Delta YS_{i,t} + \beta_{IR}\Delta IRVol_{i,t} + \beta_{CDS}\Delta CDSVol_{i,t} + \varepsilon_{i,t}.$$

Interpretation.

Table 4
Cross-sectional tests

	safe	low risk	risky
A. Dependent variable is $\Delta Yield_{i,t}$			
Intercept	0.06	−1.03	−1.99
	(0.35)	(0.81)	(1.76)
$\Delta CDS_{i,t}$	−0.13*	0.44***	0.74***
	(0.07)	(0.07)	(0.05)
$\Delta CDS_{i,t} \times \mathbb{1}_{\{\Delta DB_t \geq q(80\%)\}}$	−0.00	−0.28**	−0.31**
	(0.12)	(0.11)	(0.14)
$\Delta rf_{i,t}$	1.03***	0.77***	0.73***
	(0.02)	(0.07)	(0.09)
Observations	478	390	391
Adjusted R^2	0.88	0.49	0.57
B. Dependent variable is $\Delta CDS_{i,t}$			
Intercept	−0.09	0.25	0.59
	(0.38)	(0.84)	(1.86)
$\Delta YS_{i,t}$	−0.16**	0.35***	0.73***
	(0.07)	(0.08)	(0.08)
$\Delta IR^{Vol}_{i,t}$	0.03***	0.02	0.09***
	(0.01)	(0.01)	(0.03)
$\Delta CDS^{Vol}_{i,t}$	0.16***	0.18***	0.11***
	(0.06)	(0.04)	(0.04)
Observations	478	390	391
Adjusted R^2	0.11	0.25	0.59
Credit ratio	0.23	0.71	0.91

- For safe countries, β_{YS} is small (even negative), while volatility terms are positive and significant.
- For risky countries, β_{YS} is large and positive: CDS changes largely reflect credit-risk re-pricing.
- This is consistent with the model: regulatory demand depends on volatility and capital charges, which matter proportionally more when default risk is low.

7.10 Table 5: core sample (10 sovereigns) and regulatory proxies (Equation (12))

Equation (12) regresses CDS changes on both *credit-risk proxies* and *regulatory proxies*:

$$\Delta \mathrm{CDS}_t = \alpha + \beta_{YS}\Delta YS_t + \beta_{CtD}\Delta CtD_t + \beta_{Swptn}\Delta Swptn_t + \beta_{EDF}\Delta EDF_t + \varepsilon_t.$$

Interpretation of regressors:

- ΔYS_t: yield spread change (credit risk proxy in cash market).
- ΔCtD_t: proxy for cheapest-to-deliver option in sovereign CDS (contract-design component).
- $\Delta Swptn_t$: swaption straddle premium (interest-rate volatility; affects CVA capital via exposure option-like component).
- ΔEDF_t: dealers' expected default frequency (bank balance sheet constraint / hedging intensity proxy).

Interpretation of results (Table 5).

Table 5
Sovereign CDS premiums, credit risk, and regulatory proxies

	Interc	β^{YS}	β^{CtD}	β^{Swptn}	β^{EDF}	Adj. R^2	Credit ratio	# Obs
JAP	0.01	0.14	−0.95	0.04*	0.13*	0.09	0.11	256
	[0.02]	[0.40]	[−0.73]	[1.78]	[1.68]			
USA	−0.08	0.04	−0.15	0.00	0.05***	0.06	0.17	251
	[−0.49]	[0.36]	[−0.72]	[0.18]	[4.76]			
GER	−0.04	0.06	−1.30**	0.04***	0.16***	0.35	0.17	256
	[−0.18]	[0.91]	[−2.43]	[2.7]	[4.93]			
GBR	−0.24	0.18***	−0.54*	0.02	0.10***	0.21	0.43	256
	[−1.11]	[3.12]	[−1.88]	[1.06]	[4.76]			
FIN	0.00	0.11***	−0.20	0.01	0.14***	0.43	0.16	242
	[−0.02]	[3.32]	[−1.32]	[1.21]	[6.12]			
FRA	0.07	0.53***	0.91	0.05**	0.40***	0.57	0.51	256
	[0.16]	[7.38]	[1.41]	[1.98]	[6.65]			
AUT	−0.12	0.40***	0.67	0.03	0.27***	0.42	0.57	256
	[−0.29]	[4.43]	[1.06]	[1.2]	[4.17]			
ITA	0.04	0.51***	2.73	0.13**	0.74***	0.64	0.75	256
	[0.04]	[5.60]	[1.33]	[2.28]	[6.43]			
ESP	−0.09	0.65***	2.49*	0.08	0.40***	0.66	0.94	256
	[−0.10]	[8.11]	[1.68]	[1.23]	[4.03]			
POR	0.30	0.53***	2.24	0.15	1.09***	0.66	0.86	255
	[0.15]	[6.64]	[0.93]	[0.90]	[3.96]			

Table 6
Link between corporate bond yields and CDS premiums

	Aaa - Aa	A	Baa	Ba-C
Intercept	−0.00	−0.00	−0.00	−0.00
	(0.00)	(0.00)	(0.00)	(0.00)
ΔCDS_t	0.42***	1.02***	0.93***	0.52***
	(0.04)	(0.11)	(0.05)	(0.05)
Δrf_t	0.92***	0.89***	0.68***	0.02
	(0.02)	(0.02)	(0.04)	(0.10)
Observations	19,629	20,249	20,414	29,562
Adjusted R^2	0.42	0.34	0.35	0.30

- Safe havens: yield spreads have weak explanatory power; regulatory proxies (swaption and EDF) matter more.
- Riskier sovereigns: yield spreads dominate; credit risk explains most CDS variation.
- The "Credit ratio" summarizes how much of the full-model R^2 is attributable to the credit-risk proxies. It is low for safe havens and high for risky sovereigns, matching the model's comparative statics.

7.11 Table 6 and Table 7: corporate bond and CDS evidence

The paper tests whether the "safe-entity disconnect" also appears for *safe corporates*, and whether it depends on collateralization.

7.11.1 Table 6: by credit rating

Regression:

$$\Delta\text{Yield}_{i,t} = \alpha + \beta_{CDS}\Delta\text{CDS}_{i,t} + \beta_{rf}\Delta r^f_{i,t} + \varepsilon_{i,t}.$$

Interpretation.

Table 7
Link between bond yields and CDS premiums in different episodes

	Pre	Crisis	Post
Intercept	−0.00	0.00	0.00*
	(0.00)	(0.00)	(0.00)
ΔCDS_t	0.98***	0.48***	0.50***
	(0.09)	(0.11)	(0.07)
$\Delta CDS_t \times \mathbb{1}_{\{Corporate\}}$	−0.20	−0.10	−0.25**
	(0.27)	(0.12)	(0.11)
Δrf_t	0.90***	0.73***	0.96***
	(0.02)	(0.05)	(0.02)
Observations	36,153	12,842	19,823
Adjusted R^2	0.43	0.22	0.57

- For Aaa–Aa bonds (safest), β_{CDS} is well below one: bond yields respond less than one-for-one to CDS changes.
- For intermediate ratings (A, Baa), β_{CDS} is close to one: the standard link holds better.
- This mirrors the sovereign pattern: the disconnect is strongest where credit risk is smallest.

7.11.2 Table 7: pre-crisis vs crisis vs post-crisis, and corporate vs financial

Table 7 allows the slope on CDS to differ for nonfinancial corporates (often uncollateralized) versus financials (often collateralized), across three subperiods.

Interpretation.

- Pre-crisis: the CDS–yield link is close to one and corporate-vs-financial differences are small.
- Post-crisis: the link weakens, and the corporate interaction is negative and significant: safe nonfinancial corporates show a *stronger* disconnect than safe financials.
- This matches the CVA-capital story: the post-crisis regulatory environment and uncollateralized derivatives exposures are the relevant ingredients.

8 Summary

8.1 One-sentence

Safe sovereign CDS premiums can include a *capital-relief component* because Basel III CVA capital charges make CDS protection valuable to dealer banks even when default risk is tiny; this creates a wedge between CDS premiums and bond yield spreads that is strongest for safe entities.

8.2 Checklist of testable predictions (model → empirics)

1. CDS premiums for safe entities are driven more by regulatory variables than by credit risk.
2. CDS volumes outstanding increase with banks' uncollateralized derivatives exposures (EE proxies).
3. The CDS–bond yield link weakens for safer entities and during episodes of high dealer-bank hedging activity.

4. CDS premiums increase with proxies for $\kappa(s)$: CDS volatility, interest rate volatility, and dealers' balance-sheet constraints.

5. Similar patterns appear for safe corporates that do not collateralize derivatives (but less for financials that do).

9 Conclusion

9.1 What the paper concludes

This paper argues that a nontrivial component of sovereign (and some corporate) CDS premiums reflects *regulatory demand for capital relief* rather than only compensation for default risk. The starting point is that derivatives-dealing banks trade uncollateralized OTC derivatives with sovereigns (e.g., interest rate swaps), which creates counterparty-credit exposure and generates a *credit value adjustment (CVA)* that raises banks' risk-weighted assets and capital requirements. Because banks can reduce the CVA-related capital charge by purchasing CDS protection on the sovereign, CDS become valuable as a tool for *regulatory capital relief*, even when the sovereign is very safe.

The paper delivers a joint theoretical-and-empirical conclusion:

(i) **Theory: CDS premiums can be above "credit risk" premiums when capital relief is valuable.** In equilibrium, dealer banks' willingness to pay for CDS includes the value of lower CVA capital requirements. The seller must post initial margin when selling protection, which creates an opportunity cost that requires a positive premium even for nearly risk-free reference entities. Hence, equilibrium CDS spreads can be *significantly higher* than what default risk alone would imply, especially for safe reference entities.

(ii) **Evidence: CDS premiums and bond yield spreads disconnect for safe sovereigns.** For safe sovereigns, the paper documents that changes in bond yield spreads and changes in CDS premiums are close to unrelated. This is consistent with CDS being priced partly by a regulatory-capital channel that is not tightly linked to contemporaneous default-risk variation measured in bond yields.

(iii) **Evidence: dealer banks are net buyers and quantities line up with derivatives exposure.** Derivatives-dealing banks are net long sovereign CDS, and notional CDS amounts are related to the amount of sovereign derivatives exposure, consistent with banks using CDS to hedge CVA risk and obtain capital relief.

(iv) **Evidence: regulatory proxies explain safe-haven CDS premiums.** Variables that proxy for the strength of incentives to use CDS for capital relief (and for the cost of providing/selling CDS through margin requirements) significantly explain CDS premiums for the safest sovereigns, where pure credit-risk explanations are weakest.

(v) **External validity: a similar disconnect appears for safe corporate issuers, with cross-sectional patterns consistent with collateralization.** In corporate bond/CDS data, the paper shows a post-crisis disconnect between CDS premiums and bond yield spreads that is particularly pronounced for safe nonfinancial issuers. The disconnect is less pronounced for financial issuers, consistent with different collateralization practices and institutional details that mitigate CVA-driven hedging demand.

9.2 Why would anyone pay for CDS on a nearly risk-free entity?

At first glance, paying CDS premiums on a "safe haven" seems puzzling. The paper's intuition is that CDS provide a way to *trade and hedge a regulatory-relevant risk* that otherwise sits inside the bank's capital requirement.

1) CVA makes counterparty-credit exposure costly in capital terms. Even if a sovereign is safe, uncollateralized derivatives exposures generate CVA. Regulations require capital against CVA-related risk, so the exposure consumes scarce bank capital.

2) CDS transform CVA risk into a hedgeable, tradeable object. By buying CDS on the sovereign, the bank can reduce the CVA capital charge. Thus the bank's willingness to pay for CDS includes:

$$\text{CDS value} \approx \underbrace{\text{default-protection value}}_{\text{tiny for safe names}} + \underbrace{\text{capital-relief value}}_{\text{potentially large and first-order}}.$$

For safe havens, the second term can dominate the first.

3) Why spreads do not collapse to (near) zero for safe names. Even if default risk is close to zero, the seller of CDS typically must post initial margin. That margin ties up capital and has an opportunity cost. Therefore, sellers require a positive premium, and equilibrium spreads reflect both the buyer's capital-relief demand and the seller's margin cost.

4) Why banks do not simply issue equity instead of hedging. If Modigliani–Miller held frictionlessly, the bank could issue equity to support the extra capital requirement and avoid buying CDS. The paper argues that equity issuance is costly in practice (consistent with the logic in Froot–Stein style frameworks), so banks optimally hedge tradeable risks that affect earnings volatility and regulatory capital—and CDS make CVA risk tradeable.

5) Desk-level incentives reinforce the mechanism. Within banks, trading desks operate under risk limits and are evaluated on return on (allocated) equity capital. They therefore have direct incentives to use instruments that reduce regulatory capital consumption, including CDS used for CVA relief.

9.3 Takeaway

Safe-haven CDS spreads are not just credit-risk prices: regulatory capital relief and margin costs can generate a sizeable "safe-haven CDS premium," producing a disconnect between CDS premiums and bond yield spreads that is strongest for the safest sovereigns (and appears for safe corporates in ways consistent with collateralization differences).